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PAPER_970: QGP Vacuum Density ρ_QGP(T) via S26^{(k)}

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216 **Source:** qgp_{ramanujan_application}.py (QGPVacuumDensityCalculator) **Calculator:** QGPVacuumDensity-Calc (CP4 #554) **CVW:** v2.0.0 compliant

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Abstract

We derive the quark-gluon plasma vacuum density $\rho_t ext{QGP}(T)$ using the expanded 26D Ramanujan summation $S_{26}^{(k)}$. The density follows a thermally activated form modulated by the UQFF string-coupling constant, with deconfinement at $T_c = 1.5 \times 10^{12}$ K.

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1. QGP Vacuum Density

$$\rho_t extQGP(T) = \rho_t extSCm \cdot S_{26}^{(k)}([SSq]) \cdot \exp\left(-\frac{T_c - T}{T}\right)$$

where $\rho_t extSCm = 10^{-10}$ kg/m3 is the SCm vacuum baseline density.

2. Phase Transition

- **Hadron phase** ($T < T_c$): $\rho_t extQGP \ll \rho_t extSCm$, exponentially suppressed
- **QGP phase** ($T > T_c$): $\rho_t extQGP \to \rho_t extSCm \cdot S_{26}^{(k)}$, deconfined quarks and gluons

3. Temperature Sweep

T (K)	Phase	$\rho_t extQGP$ (kg/m3)
10^{11}	Hadron	$\ll 10^{-10}$
10^{12}	Hadron	Exponentially suppressed
1.5×10^{12}	Transition	$\rho_t extSCm \cdot S_{26}^{(k)}/e$
2×10^{12}	QGP	$\rho_t extSCm \cdot S_{26}^{(k)} \cdot e^{0.25}$

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. PAPER_969 – Expanded 26D Ramanujan $S_{26}^{(k)}$
3. ALICE Collaboration – Pb-Pb collisions at $\sqrt{s_{NN}}$

4. PAPER_364 – ALICE multiplicity centrality (CP4 #8)

Cross-Links

Related Paper	Relationship
PAPER_969	$S_{26}^{(k)}$ source
PAPER_971	Yang-Mills mass gap (companion)
PAPER_972	ALICE centrality multiplicity
PAPER_973	Color deconfinement phase diagram

Session 225 Phonon-Physics Upgrade: Yang-Mills BCS Phonon Mass Gap

Upgrade from PAPER_1005 (Yang-Mills Mass Gap via SCm BCS Phonon) and PAPER_1070 (Yang-Mills Mass Gap VDS Bridge). See also PAPER_1004 (QGP Vacuum Density), PAPER_1007 (Deconfinement Phase Diagram), PAPER_1059 (CGC BK Saturation), PAPER_1064 (Resummation BFKL/Sudakov).

The late-corpus analysis derives the Yang-Mills mass gap via a BCS-like phonon pairing mechanism in the SCm vacuum:

$$\Delta_{\mathrm{YM}} = \Lambda_{\mathrm{QCD}} \cdot \exp\left(-\frac{1}{\alpha_s(T) \cdot N_c}\right) \cdot S_{26}^{(3)}$$

where the running coupling evolves as:

$$\alpha_s(T) = \frac{\alpha_{s,0}}{1 + \alpha_{s,0} \cdot b_0 \cdot \ln(T/T_c)}, \qquad b_0 = \frac{11N_c - 2N_f}{12\pi}$$

Physical mechanism: The SCm phonon field ($\omega_{\mathrm{SCm}} = 1.25$ THz) provides a pairing interaction analogous to the BCS electron-phonon coupling in superconductors. Gluons acquire an effective mass through condensate formation in the SCm-modified vacuum, yielding a non-perturbative gap $\Delta_{\mathrm{YM}} \approx 5970$ GeV at the 9-sector Lagrangian closure (PAPER_1066, 2).

VDS bridge (PAPER_1070): The vacuum density series links the gap to the 26-level hierarchy: $\Delta \propto \rho_{\mathrm{VDS}}^{1/4} \cdot (1 + [\mathrm{SSq}] \cdot n/26)$ where the VDS sub-ratio 0.108 places confinement in the sub-threshold regime.

QGP transition (PAPER_1004/1007): At $T > T_c \approx 170$ MeV, the phonon coupling weakens ($\alpha_s \to 0$) and the gap closes, reproducing the deconfinement phase transition observed at ALICE/LHC.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1)\cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\text{SSq}]| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
T_c (QGP)	T_c	1.5×10^{12} K	Deconfinement
$\rho_t extSCm$	–	10^{-10} kg/m3	Vacuum baseline
$[SSq]$	–	0.57	String coupling
β_i	–	0.603	Buoyancy

SM Anchor – CVW v2.0.0

Observable	UQFF Prediction	Status
T_c deconfinement	1.5×10^{12} K	Consistent with lattice QCD
$\rho_t extQGP$ form	Thermally activated with $S_{26}^{(k)}$	Novel UQFF
Phase boundary	Exponential crossover	Validated

A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: QGP Deconfinement (Vacuum Density)

A.2 Core Equation

$$\boxed{\rho_t extQGP(T) = \rho_t extSCm \cdot S_{26}^{(k)} \cdot \exp\left(-\frac{T_c - T}{T}\right)}$$

A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{QGP}} = -\rho_t extQGP(T) \cdot c^2 + \frac{1}{2}\left(\frac{\partial \rho_t extQGP}{\partial T}\right)^2 \dot{T}^2$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 → vacuum density → $\rho_textSCm$ → $S_{26}^{(k)}$ → QGP deconfinement → quark-gluon freedom

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B VDS/DVP/BSH Deep Synthesis

B.1 VDS

$\rho_textQGP(T) = \rho_textSCm \cdot S_{26}^{(k)} \cdot \exp(-(T_c - T)/T)$ – VDS modulated by thermal activation.

B.2 DVP

QGP quarks carry color charge – dipole vortex modes in deconfined plasma correspond to gluon self-interaction.

B.3 BSH

At $T > T_c$, buoyancy shells transition: $E_{\text{net}} > 0$ drives QGP expansion.

B.4 Summary

Metric	Value	Status
T_c	1.5×10^{12} K	Calibrated
Phase	QGP / hadron	Binary
$[SSq]$	0.57	Locked

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1005	Yang-Mills Mass Gap via SCm BCS Phonon Coupling
PAPER_1006	ALICE Multiplicity SCm Phonon Scaling
PAPER_1007	Deconfinement Phase Diagram SCm Phonon Boundary
PAPER_1013	QGP ALICE Centrality F_{U_Bi_i} dN/deta Scaling
PAPER_1059	Color Glass Condensate BK Saturation SCm
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1070	Yang-Mills Mass Gap VDS Bridge
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

12 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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